Rotating-Diffuser Speckle as a Spatiotemporal Heating Field

A tutorial on angular intensity, temporal autocorrelation, and the spectrum $\mathcal{S}_I(\mathbf{k}, \omega)$

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Contents

1 Purpose and scope

This document develops a quantitative model for light scattered from an in-plane rotating diffuser. The intended application is a stochastic optical heating field for thermal-transport experiments.

The primary experimental observables are:

1. The mean scattered intensity as a function of optical scattering vector,
$$\bar{I}(Q). \tag{1}$$
2. The normalized temporal intensity autocorrelation at a selected scattering vector,
$$g_2(Q,\tau). \tag{2}$$
3. The spatial and temporal power spectral density of the heating intensity in the sample plane,
$$\mathcal{S}_I(\mathbf{k},\omega). \tag{3}$$

These quantities are related, but they are not interchangeable.

In particular, knowledge of $\bar{I}(Q)$ and a collection of local temporal autocorrelations does not, in general, determine the complete spatiotemporal spectrum $\mathcal{S}_I(\mathbf{k},\omega)$. The missing information is the correlation between different spatial positions or different optical modes.

The ideal model developed first assumes:

- monochromatic coherent illumination;
- a thin random diffuser;
- a circular illuminated area of diameter D;
- illumination centered on the diffuser rotation axis;
- uniform illumination across that circular area;
- observation in the far field or a true Fourier plane;
- a point detector that resolves one speckle or fewer;
- rigid rotation of the far-field speckle pattern;
- statistically fully developed speckle.

Later sections discuss off-axis illumination, finite detector area, nonuniform beams, and the distinction between rigid rotation and speckle boiling.

2 Geometry and notation

Let the optical wavelength be λ, with optical wavenumber

$$k_0 = \frac{2\pi}{\lambda}. \tag{4}$$

The diffuser rotates in its own plane with angular velocity

$$\Omega = 2\pi f_{\mathrm{rot}}, \tag{5}$$

where f_{rot} is the rotation frequency in revolutions per second.

The incident optical wavevector is $\mathbf{k}_{\mathrm{i}}$ and the scattered wavevector is $\mathbf{k}_{\mathrm{s}}$. The optical scattering vector is

$$\mathbf{Q} = \mathbf{k}_{\mathrm{s}} - \mathbf{k}_{\mathrm{i}}. \tag{6}$$

For elastic scattering,

$$Q = |\mathbf{Q}| = 2k_0 \sin\left(\frac{\theta}{2}\right), \tag{7}$$

where θ is the angle between the incident and scattered beams.

The transverse optical wavevector in a Fourier plane is

$$Q_\perp = k_0 \sin\theta. \tag{8}$$

For small angles,

$$Q \approx Q_\perp \approx k_0\theta. \tag{9}$$

The thermal-heating spatial frequency will be denoted by $\mathbf{k}$. This is not the same object as the optical scattering vector $\mathbf{Q}$.

> **Notation warning.** Q describes the direction into which the optical field is scattered. The vector $\mathbf{k}$ describes spatial Fourier components of the optical *intensity* pattern that heats the sample.

3 Suggested optical arrangement

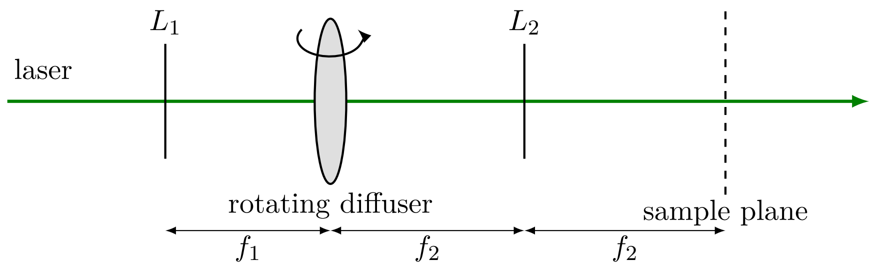
A simple Fourier-plane arrangement is

$$\text{laser} \to L_1 \to \text{rotating diffuser} \to L_2 \to \text{sample plane}. \tag{10}$$

A nominal focal arrangement is

$$d(L_1, \text{diffuser}) = f_1, \qquad d(\text{diffuser}, L_2) = f_2, \qquad d(L_2, \text{sample plane}) = f_2. \tag{11}$$

A basic schematic is shown below.



If the field at the sample plane is later relayed onto a camera, the relay optics must be placed downstream of the defined sample plane. Changing the relay magnification does not change the heating field at the sample, provided the relay does not feed light back into the illumination system.

4 Mean angular intensity $\overline{I}(Q)$

4.1 Definition

The angular scattering envelope is

$$\overline{I}(Q) = \langle I(Q,t) \rangle_t \, . \tag{12}$$

Experimentally, one may rotate a point detector around the diffuser or place the detector at successive positions in a Fourier plane.

If the detector is at a physical distance z from the diffuser and is displaced laterally by ρ, then

$$\theta = \tan^{-1}\left(\frac{\rho}{z}\right). \tag{13}$$

The full optical scattering-vector magnitude is

$$Q(\rho) = 2k_0 \sin\left[\frac{1}{2}\tan^{-1}\left(\frac{\rho}{z}\right)\right]. \tag{14}$$

In a Fourier plane formed by a lens of focal length f,

$$\rho = f \tan\theta, \tag{15}$$

and therefore

$$Q(\rho) = 2k_0 \sin\left[\frac{1}{2}\tan^{-1}\left(\frac{\rho}{f}\right)\right]. \tag{16}$$

For the transverse optical wavevector,

$$Q_\perp = k_0 \sin\theta = k_0 \frac{\rho}{\sqrt{f^2+\rho^2}}. \tag{17}$$

For $\rho \ll f$,

$$Q_\perp \approx k_0 \frac{\rho}{f}. \tag{18}$$

4.2 Count-rate normalization

Suppose the point detector records a mean count rate $\overline{\mathcal{R}}(\theta)$. Correct this for:

- dark-count rate $\mathcal{R}_{\rm dark}$;
- detector quantum efficiency $\eta(\lambda)$;
- detector projected area;
- collection solid angle;
- polarization-dependent response;
- neutral-density filters;
- dead-time losses, if significant.

A relative angular intensity may be estimated as

$$\overline{I}(Q) \propto \frac{\overline{\mathcal{R}}(Q) - \mathcal{R}_{\rm dark}}{\eta(\lambda)\,\Delta\Omega(Q)}. \tag{19}$$

For a small circular detector of active diameter $d_{\rm det}$ at distance z,

$$\Delta\Omega \approx \frac{\pi d_{\rm det}^2/4}{z^2}. \tag{20}$$

At large observation angles, the projected detector area can introduce an additional cosine factor. It is safer to calculate the collection geometry explicitly.

4.3 Role of the 600-grit specification

The nominal grit affects the angular scattering envelope $\overline{I}(Q)$ and the microscopic statistics of the diffuser surface.

It does not appear explicitly in the ideal rigid-rotation correlation formula derived below. In that model, the short-delay correlation width is set principally by:

$$D, \quad \lambda, \quad \Omega, \quad \theta. \tag{21}$$

The actual diffuser can introduce additional boiling or loss of correlation, which must be measured.

5 Temporal intensity autocorrelation

5.1 Definition of g_2

At a fixed detector coordinate or scattering angle, define

$$g_2(Q,\tau) = \frac{\langle I(Q,t)I(Q,t+\tau)\rangle_t}{\langle I(Q,t)\rangle_t^2}. \tag{22}$$

Define the intensity fluctuation

$$\delta I(Q,t) = I(Q,t) - \overline{I}(Q). \tag{23}$$

The unnormalized covariance is

$$C_I(Q,\tau) = \langle \delta I(Q,t)\delta I(Q,t+\tau)\rangle\,. \tag{24}$$

The two are related by

$$C_I(Q,\tau) = \overline{I}(Q)^2\left[g_2(Q,\tau) - 1\right]. \tag{25}$$

At zero delay,

$$g_2(Q,0) - 1 = \beta(Q), \tag{26}$$

where β is the measured speckle contrast.

For an ideal single-mode, single-polarization detector,

$$\beta \to 1. \tag{27}$$

Finite detector area, polarization mixing, finite time bins, and imperfect spatial coherence reduce β.

6 Ideal centered-rotation model

6.1 Physical picture

Suppose a circular area of diameter D is illuminated symmetrically about the rotation axis. In the ideal Fourier-plane limit, the far-field speckle pattern rotates rigidly with the diffuser.

At scattering angle θ, two detector samples separated by diffuser rotation angle

$$\Delta\phi = \Omega\tau \tag{28}$$

correspond to two points on a circle in transverse optical-wavevector space.

Their transverse separation is

$$\Delta Q_\perp = 2k_0 \sin\theta \sin\left(\frac{\Delta\phi}{2}\right). \tag{29}$$

For a uniformly illuminated circular aperture, the normalized field correlation is the familiar jinc function,

$$\gamma_E = \frac{2J_1(X)}{X}, \tag{30}$$

where

$$X = \frac{D}{2}\Delta Q_\perp. \tag{31}$$

Substituting Eq. (29),

$$\boxed{X(\theta,\tau) = \frac{2\pi D}{\lambda}\sin\theta\sin\left(\frac{\Omega\tau}{2}\right)}. \tag{32}$$

For fully developed Gaussian speckle, the Siegert relation gives

$$\boxed{g_2(\theta,\tau) = 1 + \beta\left[\frac{2J_1(X)}{X}\right]^2}. \tag{33}$$

The limit at $X = 0$ is understood to be

$$\lim_{X\to 0}\frac{2J_1(X)}{X} = 1. \tag{34}$$

6.2 Short-delay form

When

$$\Omega\tau \ll 1, \tag{35}$$

then

$$\sin\left(\frac{\Omega\tau}{2}\right) \approx \frac{\Omega\tau}{2}. \tag{36}$$

Equation (32) becomes

$$X \approx \frac{\pi D\Omega}{\lambda}\sin\theta\,\tau. \tag{37}$$

Therefore,

$$\boxed{g_2(\theta,\tau) - 1 \approx \beta\left[\frac{2J_1\left(\dfrac{\pi D\Omega}{\lambda}\sin\theta\,\tau\right)}{\dfrac{\pi D\Omega}{\lambda}\sin\theta\,\tau}\right]^2}. \tag{38}$$

This curve is not exponential. It has an Airy-like central lobe and weak oscillatory sidelobes.

7 Correlation-time estimates

7.1 $1/e$ correlation time

Define $\tau_{1/e}$ by

$$g_2(\tau_{1/e}) - 1 = \frac{\beta}{e}. \tag{39}$$

The numerical solution of

$$\left[\frac{2J_1(X_{1/e})}{X_{1/e}}\right]^2 = e^{-1} \tag{40}$$

is

$$X_{1/e} = 1.91499. \tag{41}$$

Using Eq. (37),

$$\boxed{\tau_{1/e} = \frac{1.91499\,\lambda}{\pi D\Omega \sin\theta}}. \tag{42}$$

7.2 First zero

The first zero of $J_1(X)$ occurs at

$$X_0 = 3.83171. \tag{43}$$

The first zero of the ideal short-delay correlation is therefore

$$\boxed{\tau_0 = \frac{3.83171\,\lambda}{\pi D\Omega \sin\theta}}. \tag{44}$$

7.3 Scaling laws

Equations (42) and (44) imply

$$\tau_c \propto \frac{\lambda}{D\Omega \sin\theta}. \tag{45}$$

Thus:

- increasing illuminated diameter produces smaller far-field speckles and shorter point-detector correlation times;
- increasing rotation speed shortens the correlation time linearly;
- increasing detector angle shortens the correlation time through $\sin\theta$;
- at exact forward scattering, $\theta = 0$, ideal rotation causes no decorrelation.

At $\theta = 0$,

$$X = 0 \tag{46}$$

for every delay, and therefore

$$g_2(0,\tau) = 1 + \beta. \tag{47}$$

Any decay measured at exact forward scattering must come from nonideal effects such as wobble, imperfect centering, amplitude variation, diffuser boiling, vibration, or beam-pointing motion.

8 Numerical example

Take

$$\lambda = 532\,\text{nm}, \tag{48}$$
$$D = 25\,\text{mm}, \tag{49}$$
$$f_{\text{rot}} = 1\,\text{Hz}, \tag{50}$$
$$\Omega = 2\pi\ \text{rad/s}. \tag{51}$$

Equation (42) becomes

$$\boxed{\tau_{1/e} = \frac{2.064\,\mu\text{s}}{\sin\theta}}. \tag{52}$$

Equation (44) becomes

$$\boxed{\tau_0 = \frac{4.131\,\mu\text{s}}{\sin\theta}}. \tag{53}$$

Table 1: Ideal centered-illumination correlation times for a 25 mm illuminated diameter, 532 nm wavelength, and one revolution per second.

θ (degrees)	$\tau_{1/e}$	τ_0
0.01	11.8 ms	23.7 ms
0.05	2.37 ms	4.73 ms
0.10	1.18 ms	2.37 ms
0.50	237 µs	473 µs
1.00	118 µs	237 µs
5.00	23.7 µs	47.4 µs
10.0	11.9 µs	23.8 µs
30.0	4.13 µs	8.26 µs
60.0	2.38 µs	4.77 µs

The complete correlation is periodic because the same diffuser orientation returns after every revolution:

$$g_2\left(\tau + \frac{2\pi}{\Omega}\right) = g_2(\tau). \tag{54}$$

For one revolution per second,

$$g_2(\tau + 1\ \text{s}) = g_2(\tau). \tag{55}$$

The experiment should therefore show narrow correlation echoes at integer multiples of the rotation period, provided the diffuser and optical alignment remain stable.

9 Finite detector size

A detector with aperture function $A_{\text{det}}(\mathbf{r})$ measures

$$I_{\text{det}}(t) = \int A_{\text{det}}(\mathbf{r} - \mathbf{r}_d) I(\mathbf{r}, t) \mathrm{d}^2 r. \tag{56}$$

If the detector integrates over N statistically independent speckles, the contrast is approximately

$$\beta \sim \frac{1}{N}. \tag{57}$$

The correlation width may also broaden slightly because of spatial averaging.

For free-space propagation over distance z, a characteristic transverse speckle size is

$$\xi_{\mathrm{sp}} \sim \frac{\lambda z}{D}. \tag{58}$$

In a Fourier plane of a lens with focal length f,

$$\xi_{\mathrm{sp}} \sim \frac{\lambda f}{D}. \tag{59}$$

A detector of active diameter d_{det} approximately resolves one speckle when

$$d_{\mathrm{det}} \lesssim \xi_{\mathrm{sp}}. \tag{60}$$

For a 20 µm detector, the required Fourier-plane focal length is roughly

$$f \gtrsim \frac{d_{\mathrm{det}} D}{\lambda}. \tag{61}$$

For $D = 25\,\mathrm{mm}$ and $\lambda = 532\,\mathrm{nm}$,

$$f \gtrsim 0.94\,\mathrm{m}. \tag{62}$$

This estimate should be checked experimentally because definitions of speckle diameter differ by factors of order unity.

10 Off-axis illumination

Let the beam center be displaced by radius R from the diffuser rotation axis. The local diffuser-surface speed is

$$v_d = \Omega R. \tag{63}$$

Off-axis illumination can produce a combination of:

- rigid speckle rotation;
- lateral speckle translation;
- shear;
- speckle boiling;
- loss of exact recurrence.

There is no universal expression depending only on R, θ, and Ω. The result also depends on:

$$D, \quad \lambda, \quad \text{beam profile}, \quad \text{diffuser statistics}, \quad \text{ABCD optical system}, \quad \text{detector aperture}. \tag{64}$$

A practical phenomenological extension is

$$\boxed{g_2(\tau; R, \theta, \Omega) - 1 = \beta \left[\frac{2J_1(X)}{X}\right]^2 B(\Omega R \tau)} \tag{65}$$

with X given by Eq. (32).

A simple Gaussian model is

$$B(\Omega R\tau) = \exp\left[-\left(\frac{\Omega R\tau}{\ell_R}\right)^2\right], \tag{66}$$

so that

$$\boxed{g_2 - 1 = \beta\left[\frac{2J_1(X)}{X}\right]^2 \exp\left[-\left(\frac{\Omega R\tau}{\ell_R}\right)^2\right].} \tag{67}$$

Here ℓ_R is an effective diffuser-displacement decorrelation length. It must be measured rather than inferred reliably from the nominal grit.

If the additional decorrelation dominates, the corresponding time is

$$\tau_R \sim \frac{\ell_R}{\Omega R}. \tag{68}$$

A useful experimental test is therefore

$$\tau_R^{-1} \propto \Omega R. \tag{69}$$

The inferred displacement scale is

$$\boxed{\ell_R = \Omega R\tau_R.} \tag{70}$$

11 Temporal spectrum from the measured autocorrelation

The Wiener–Khinchin relation connects the temporal covariance and temporal power spectral density.

At fixed Q,

$$S_I(Q,\omega) = \int_{-\infty}^{\infty} C_I(Q,\tau)e^{-\mathrm{i}\omega\tau}\mathrm{d}\tau. \tag{71}$$

Using Eq. (25),

$$\boxed{S_I(Q,\omega) = \overline{I}(Q)^2 \int_{-\infty}^{\infty} \left[g_2(Q,\tau) - 1\right] e^{-\mathrm{i}\omega\tau}\mathrm{d}\tau.} \tag{72}$$

Thus, the required ingredients at each optical scattering vector are:

$$\overline{I}(Q) \quad \text{and} \quad g_2(Q,\tau). \tag{73}$$

The normalized g_2 alone gives the spectral shape but not its absolute power. The factor $\overline{I}(Q)^2$ restores the covariance scale.

For a real stationary signal,

$$C_I(Q,-\tau) = C_I(Q,\tau), \tag{74}$$

so

$$S_I(Q,\omega) = 2\int_0^{\infty} C_I(Q,\tau)\cos(\omega\tau)\mathrm{d}\tau. \tag{75}$$

11.1 Discrete FPGA data

Suppose the correlator returns delays

$$\tau_n = n\Delta t \tag{76}$$

on a uniform grid. Compute

$$C_n = \overline{I}^2 \left[g_2(\tau_n) - 1\right]. \tag{77}$$

A discrete cosine transform gives

$$S_I(\omega_m) \approx 2\Delta t \sum_{n=0}^{N-1} w_n C_n \cos(\omega_m \tau_n), \tag{78}$$

where w_n is an optional window.

For multi-tau lag spacing, the points are not uniformly spaced. Suitable approaches are:

1. interpolate $C_I(\tau)$ onto a uniform grid;
2. use a nonuniform Fourier transform;
3. fit a parametric model to g_2 and transform the fitted function;
4. numerically integrate the cosine transform using the actual lag intervals.

The periodic recurrence at the rotation frequency produces discrete spectral structure at harmonics of Ω when sufficiently long acquisitions are included.

12 The full spatial-temporal intensity spectrum

12.1 Definition

Let the sample-plane intensity be

$$I(\mathbf{r}, t) = \overline{I}(\mathbf{r}) + \delta I(\mathbf{r}, t). \tag{79}$$

For a statistically homogeneous field, define

$$C_I(\Delta\mathbf{r}, \tau) = \langle \delta I(\mathbf{r}, t)\delta I(\mathbf{r} + \Delta\mathbf{r}, t + \tau)\rangle. \tag{80}$$

The spatiotemporal power spectral density is

$$\boxed{\mathcal{S}_I(\mathbf{k}, \omega) = \int\int C_I(\Delta\mathbf{r}, \tau) e^{-\mathrm{i}(\mathbf{k}\cdot\Delta\mathbf{r} - \omega\tau)} \mathrm{d}^2(\Delta r)\, \mathrm{d}\tau.} \tag{81}$$

Equivalently,

$$C_I(\Delta\mathbf{r}, \tau) = \frac{1}{(2\pi)^3} \int\int \mathcal{S}_I(\mathbf{k}, \omega) e^{\mathrm{i}(\mathbf{k}\cdot\Delta\mathbf{r} - \omega\tau)} \mathrm{d}^2k\, \mathrm{d}\omega. \tag{82}$$

The units of $\mathbf{k}$ are inverse length. The units of ω are radians per second.

If cyclic spatial and temporal frequencies are preferred, define

$$\boldsymbol{\nu} = \frac{\mathbf{k}}{2\pi}, \qquad f = \frac{\omega}{2\pi}. \tag{83}$$

Then $\boldsymbol{\nu}$ has units of cycles per unit length and f has units of hertz.

13 Why $\overline{I}(Q)$ is not enough

The optical angular envelope $\overline{I}(Q)$ describes the average optical power in each propagation direction.

The sample-plane optical field can be written as an angular-spectrum superposition,

$$E(\mathbf{r},t) = \int A(\boldsymbol{\kappa},t)e^{\mathrm{i}\boldsymbol{\kappa}\cdot\mathbf{r}}\mathrm{d}^2\kappa. \tag{84}$$

Here $\boldsymbol{\kappa}$ is an optical transverse wavevector.

The optical intensity is

$$I(\mathbf{r},t) = E(\mathbf{r},t)E^*(\mathbf{r},t). \tag{85}$$

Its spatial Fourier transform is

$$\boxed{\widetilde{I}(\mathbf{k},t) = \int A(\boldsymbol{\kappa},t)A^*(\boldsymbol{\kappa}-\mathbf{k},t)\mathrm{d}^2\kappa.} \tag{86}$$

Thus the heating-intensity spatial spectrum is produced by differences between pairs of optical wavevectors.

The mean angular intensity gives

$$\overline{I}(Q) \propto \left\langle |A(\boldsymbol{\kappa},t)|^2 \right\rangle, \tag{87}$$

but Eq. (86) also depends on relative phases and cross-correlations between different optical modes.

Therefore,

$$\boxed{\overline{I}(Q) \text{ alone does not determine } \mathcal{S}_I(\mathbf{k},\omega).} \tag{88}$$

Similarly, measuring $g_2(Q,\tau)$ at isolated detector positions gives local temporal spectra but does not reveal correlations between different spatial locations.

14 Ways to obtain $\mathcal{S}_I(\mathbf{k},\omega)$

14.1 Method 1: Camera movie at the sample plane

This is the most direct method.

Record

$$I(x,y,t_n) \tag{89}$$

at the actual sample plane or at a calibrated conjugate image plane.

Subtract the temporal mean:

$$\delta I(x,y,t) = I(x,y,t) - \langle I(x,y,t)\rangle_t\,. \tag{90}$$

Compute the three-dimensional Fourier transform:

$$\widetilde{\delta I}(k_x,k_y,\omega) = \iiint \delta I(x,y,t)e^{-\mathrm{i}(k_x x + k_y y - \omega t)}\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}t. \tag{91}$$

Then estimate

$$\boxed{\mathcal{S}_I(k_x, k_y, \omega) = \frac{1}{AT} \left| \widetilde{\delta I}(k_x, k_y, \omega) \right|^2.} \tag{92}$$

Here A is the analyzed image area and T is the acquisition duration.
In practice:

1. subtract the camera dark frame;
2. correct pixel gain if necessary;
3. subtract the time-averaged image;
4. apply spatial and temporal windows;
5. calculate a three-dimensional FFT;
6. square the magnitude;
7. average neighboring blocks or repeated acquisitions.

14.2 Method 2: Two-point detector correlations

With one detector fixed at $\mathbf{r}_0$ and one detector at $\mathbf{r}$, measure

$$C_I(\mathbf{r}_0, \mathbf{r}, \tau) = \langle \delta I(\mathbf{r}_0, t) \delta I(\mathbf{r}, t + \tau) \rangle . \tag{93}$$

If the field is homogeneous,

$$C_I(\mathbf{r}_0, \mathbf{r}, \tau) = C_I(\mathbf{r} - \mathbf{r}_0, \tau). \tag{94}$$

Scanning $\mathbf{r}$ reconstructs Eq. (80), whose Fourier transform gives $\mathcal{S}_I(\mathbf{k}, \omega)$.
A set of single-point autocorrelations measures only

$$C_I(\mathbf{r}, \mathbf{r}, \tau). \tag{95}$$

That is the diagonal of the full spatial covariance matrix and is insufficient for general reconstruction.

14.3 Method 3: Rigid-rotation model

If the pattern is known to rotate rigidly,

$$I(\mathbf{r}, t) = I_0 \left(\mathbf{R}_{-\Omega t} \mathbf{r} \right), \tag{96}$$

where $\mathbf{R}_\phi$ is a two-dimensional rotation matrix.
One static spatial image $I_0(x, y)$ then predicts the entire movie.
A synthetic sequence may be generated by rotating the measured frame through

$$\phi_n = \Omega t_n. \tag{97}$$

The resulting synthetic movie can be transformed using Eq. (92).
This method is valid only if experiments show:

- correlation echoes at every revolution;
- correlation times scaling as $1/\Omega$;
- minimal speckle boiling;
- minimal vibration and wobble;
- reproducible spatial patterns after each revolution.

15 Structure of the spectrum for rigid rotation

Express a static pattern in polar coordinates as an angular Fourier series:

$$I_0(r,\phi) = \sum_{m=-\infty}^{\infty} I_m(r)e^{\mathrm{i}m\phi}. \tag{98}$$

Under rigid rotation,

$$I(r,\phi,t) = I_0(r,\phi-\Omega t). \tag{99}$$

Therefore,

$$I(r,\phi,t) = \sum_m I_m(r)e^{\mathrm{i}m\phi}e^{-\mathrm{i}m\Omega t}. \tag{100}$$

The temporal frequencies are discrete:

$$\boxed{\omega_m = m\Omega.} \tag{101}$$

Thus a perfectly repeating rotating pattern is not temporally white. Its spectrum consists of harmonics of the rotation frequency, with amplitudes set by the angular spatial structure.

A finite observation time broadens the spectral lines. Mechanical jitter, surface nonuniformity, and speckle boiling can produce continuous spectral components around the harmonics.

This is an important conceptual result:

$$\boxed{\text{Rigid rotation couples spatial angular order } m \text{ to temporal frequency } m\Omega.} \tag{102}$$

It does not generate independent spatial and temporal frequencies.

16 Local traveling-pattern approximation

Over a small region, a rotating pattern can resemble translation:

$$I(\mathbf{r},t) \approx I_0(\mathbf{r}-\mathbf{v}t). \tag{103}$$

The Fourier representation then satisfies

$$\widetilde{I}(\mathbf{k},\omega) \propto \widetilde{I}_0(\mathbf{k})\delta(\omega-\mathbf{k}\cdot\mathbf{v}). \tag{104}$$

The intensity power is concentrated along the ridge

$$\boxed{\omega = \mathbf{k}\cdot\mathbf{v}.} \tag{105}$$

A measured $\mathcal{S}_I(\mathbf{k}, \omega)$ with narrow ridges indicates traveling or advected speckle. A broad distribution at fixed $\mathbf{k}$ indicates boiling, deformation, or additional stochastic evolution.

17 From optical intensity to absorbed heating

Let the sample absorptance be $A_{\text{abs}}(\mathbf{r})$.

For surface-localized absorption,

$$P_{\text{abs}}(\mathbf{r}, t) = A_{\text{abs}}(\mathbf{r}) I(\mathbf{r}, t). \tag{106}$$

For spatially uniform absorptance,

$$\delta P_{\text{abs}}(\mathbf{r}, t) = A_{\text{abs}} \delta I(\mathbf{r}, t). \tag{107}$$

The absorbed-power spectrum is then

$$\mathcal{S}_P(\mathbf{k}, \omega) = A_{\text{abs}}^2 \mathcal{S}_I(\mathbf{k}, \omega). \tag{108}$$

If absorption occurs over depth z with coefficient μ_a,

$$P_{\text{abs}}(\mathbf{r}, z, t) = \mu_a I(\mathbf{r}, t) e^{-\mu_a z}. \tag{109}$$

The three-dimensional heating spectrum contains the depth transform

$$\widetilde{P}_{\text{abs}}(\mathbf{k}, k_z, \omega) = \widetilde{I}(\mathbf{k}, \omega) \frac{\mu_a}{\mu_a + \mathrm{i} k_z}. \tag{110}$$

18 Connection to thermal transport

For an isotropic diffusive material with volumetric heat capacity C and thermal conductivity κ, the Fourier-domain heat equation is

$$\left[\kappa k^2 - \mathrm{i}\omega C\right] \widetilde{T}(\mathbf{k}, \omega) = \widetilde{P}_{\text{abs}}(\mathbf{k}, \omega). \tag{111}$$

Therefore,

$$\boxed{\widetilde{T}(\mathbf{k}, \omega) = H_{\text{th}}(\mathbf{k}, \omega) \widetilde{P}_{\text{abs}}(\mathbf{k}, \omega),} \tag{112}$$

where

$$\boxed{H_{\text{th}}(\mathbf{k}, \omega) = \frac{1}{\kappa k^2 - \mathrm{i}\omega C}.} \tag{113}$$

With thermal diffusivity

$$\alpha = \frac{\kappa}{C}, \tag{114}$$

the characteristic diffusive relaxation rate is

$$\omega_{\text{th}} = \alpha k^2. \tag{115}$$

The corresponding time is

$$\tau_{\mathrm{th}} = \frac{1}{\alpha k^2}. \tag{116}$$

If the spatial period is

$$\Lambda = \frac{2\pi}{k}, \tag{117}$$

then

$$\tau_{\mathrm{th}} = \frac{\Lambda^2}{4\pi^2\alpha}. \tag{118}$$

A useful stochastic thermal experiment requires appreciable input power in the region of (k, ω) where the sample response is sensitive to the transport physics of interest.

19 Recommended experimental workflow

19.1 Stage 1: Mean scattering envelope

At fixed diffuser speed, measure

$$\bar{I}(Q) \tag{119}$$

over the usable angular range.

Record:

- detector angle;
- detector distance or Fourier-plane focal length;
- count rate;
- dark count;
- neutral-density attenuation;
- detector aperture;
- polarization;
- diffuser radius and illuminated diameter.

19.2 Stage 2: Centered temporal correlations

Center the illuminated circular area on the diffuser rotation axis.

Measure

$$g_2(\theta, \tau) \tag{120}$$

at several angles and diffuser speeds.

Test the collapse

$$g_2(\theta, \tau; \Omega) = F\left(\Omega\tau \sin\theta\right). \tag{121}$$

Test the predicted scaling

$$\tau_{1/e}^{-1} \propto D\Omega \sin\theta. \tag{122}$$

19.3 Stage 3: Rotation echoes

Acquire correlations through at least several rotation periods.

Check whether

$$g_2(nT_{\mathrm{rot}}) \approx 1 + \beta, \tag{123}$$

where

$$T_{\mathrm{rot}} = \frac{2\pi}{\Omega}. \tag{124}$$

Loss of echo amplitude with increasing n quantifies long-term drift or nonrepeatability.

19.4 Stage 4: Off-axis illumination

Repeat for several offsets R.

Fit Eq. (67) or another empirical decorrelation model.

Test

$$\tau_R^{-1} \propto \Omega R. \tag{125}$$

Determine

$$\ell_R = \Omega R \tau_R. \tag{126}$$

19.5 Stage 5: Spatially resolved measurement

Place a camera at the actual sample plane or a calibrated conjugate plane.

Use a diffuser speed low enough to avoid temporal aliasing.

Measure

$$I(x, y, t) \tag{127}$$

and calculate

$$\mathcal{S}_I(k_x, k_y, \omega). \tag{128}$$

Then increase the speed and verify that the FPGA-measured temporal correlations scale consistently with Ω.

19.6 Stage 6: Thermal experiment

Replace the camera with the sample without changing the upstream heating optics.

The measured heating spectrum should be treated as

$$\mathcal{S}_P(\mathbf{k}, \omega) = A_{\mathrm{abs}}^2 \mathcal{S}_I(\mathbf{k}, \omega), \tag{129}$$

subject to corrections for reflection, sample roughness, nonuniform absorptance, and finite penetration depth.

20 Data-analysis pseudocode

20.1 Temporal spectrum from FPGA correlation data

Listing 1: MATLAB/Octave-style pseudocode for a temporal spectrum.

```
% Inputs:
% tau        delay times [s]
% g2         normalized intensity autocorrelation
% Imean      mean intensity or mean count rate
%
% First obtain the covariance:
C = Imean.^2 .* (g2 - 1);

% For multi-tau data, interpolate onto a uniform lag grid:
tau_uniform = linspace(min(tau), max(tau), 2^nextpow2(length(tau)));
C_uniform = interp1(tau, C, tau_uniform, 'pchip', 0);

% Optional window:
w = hann(length(C_uniform))';
Cw = C_uniform .* w;

% Construct an even covariance for a two-sided FFT:
C_two_sided = [fliplr(Cw(2:end)), Cw];

dtau = tau_uniform(2) - tau_uniform(1);
S = real(fftshift(fft(ifftshift(C_two_sided)))) * dtau;

N = length(C_two_sided);
omega = 2*pi*((-floor(N/2)):(ceil(N/2)-1))/(N*dtau);

% Retain nonnegative frequencies if desired:
keep = omega >= 0;
omega_positive = omega(keep);
S_positive = S(keep);
```

20.2 Three-dimensional spectrum from a camera movie

Listing 2: MATLAB/Octave-style pseudocode for $\mathcal{S}_I(k_x, k_y, \omega)$.

```
% movie has dimensions Ny x Nx x Nt
% dx, dy are sample-plane pixel spacings
% dt is frame spacing

movie = double(movie);

% Subtract the time-averaged illumination:
mean_image = mean(movie, 3);
for n = 1:size(movie,3)
    movie(:,:,n) = movie(:,:,n) - mean_image;
end

% Apply separable windows:
```

```
wx = hann(size(movie,2))';
wy = hann(size(movie,1));
wt = reshape(hann(size(movie,3)), 1, 1, []);

window3 = reshape(wy, [], 1, 1) ...
        .* reshape(wx, 1, [], 1) ...
        .* wt;

movie_w = movie .* window3;

% Three-dimensional Fourier transform:
F = fftshift(fftn(movie_w));

% Periodogram estimate:
A = size(movie,1)*dy * size(movie,2)*dx;
T = size(movie,3)*dt;
S_I = abs(F).^2 / (A*T);

% Coordinate axes:
Ny = size(movie,1);
Nx = size(movie,2);
Nt = size(movie,3);

kx = 2*pi*((-floor(Nx/2)):(ceil(Nx/2)-1))/(Nx*dx);
ky = 2*pi*((-floor(Ny/2)):(ceil(Ny/2)-1))/(Ny*dy);
omega = 2*pi*((-floor(Nt/2)):(ceil(Nt/2)-1))/(Nt*dt);
```

21 Interpretive checks

The following signatures distinguish several regimes.

21.1 Rigid rotation

Expected observations:

$$\tau_c^{-1} \propto \Omega \sin\theta, \tag{130}$$

strong recurrence at each rotation period, and spectral power concentrated at harmonics

$$\omega = m\Omega. \tag{131}$$

21.2 Local translation

Expected observations:

$$\omega = \mathbf{k} \cdot \mathbf{v} \tag{132}$$

in the measured $\mathcal{S}_I(\mathbf{k}, \omega)$.

The two-point correlation peak shifts according to

$$\tau_{\text{peak}} \approx \frac{\Delta\mathbf{r} \cdot \hat{\mathbf{v}}}{v}. \tag{133}$$

21.3 Speckle boiling

Expected observations include:

- loss of complete recurrence after one revolution;
- continuous temporal broadening around rotation harmonics;
- faster decay than predicted by Eq. (33);
- weak or absent displacement-dependent correlation peaks.

21.4 Mechanical wobble or beam motion

Expected observations include:

- modulation at the fundamental rotation frequency;
- decay at $\theta = 0$;
- position-dependent mean intensity;
- nonreproducible echo amplitude;
- correlations that do not collapse against $\Omega\tau$.

22 Main conclusions

For centered illumination of a uniformly illuminated circular diffuser in a Fourier-plane geometry, the ideal temporal autocorrelation is

$$g_2(\theta, \tau) = 1 + \beta \left[\frac{2J_1\left(\frac{2\pi D}{\lambda} \sin\theta \sin\frac{\Omega\tau}{2}\right)}{\frac{2\pi D}{\lambda} \sin\theta \sin\frac{\Omega\tau}{2}} \right]^2 . \tag{134}$$

For short delays,

$$\tau_{1/e} = \frac{1.91499\,\lambda}{\pi D \Omega \sin\theta}. \tag{135}$$

This expression reproduces the microsecond correlation-time table for a 25 mm illuminated diameter at one revolution per second.

The angular mean intensity $\overline{I}(Q)$ can be measured by detector-angle scanning. At each Q, the temporal spectrum follows from

$$S_I(Q, \omega) = \overline{I}(Q)^2 \mathcal{F}_{\tau\to\omega} \left\{ g_2(Q, \tau) - 1 \right\}. \tag{136}$$

However,

$$\overline{I}(Q) \quad \text{and} \quad g_2(Q, \tau) \tag{137}$$

do not generally determine the complete sample-plane spectrum

$$\mathcal{S}_I(\mathbf{k}, \omega). \tag{138}$$

That quantity requires a camera movie, two-point spatial correlations, or a validated dynamical model such as rigid rotation.

A rigidly rotating pattern produces strong coupling between spatial and temporal structure rather than independent broadband noise. In angular harmonics,

$$\omega = m\Omega. \tag{139}$$

Locally translating speckle produces

$$\omega = \mathbf{k} \cdot \mathbf{v}. \tag{140}$$

These relations should be measured before interpreting the diffuser as a spatiotemporally broadband heating source.

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